Dual-Indexed Local Jacobian Scanning and Structural Auditing of Internal Targets in Vision Transformers

Abstract

A scalar Neural Network Scanner (NNS) learning image equals same-target Gradient × Input under exact current-point derivatives without additional redistribution. We define local maps that retain input coordinates and concrete consumer uses, separating structurally valid cuts, pointwise differential closure, and attention-branch differential partitions. Structural audits combine closure checks with an independent CPU float64 analytic reference and prescribed wrong-slot and fan-out faults. A 100-image unit subset and 400-image structural analysis of two fixed checkpoints show distinct block-index trajectories of normalized-input Endpoint spatial mass. Three-coordinate checks retain the main directions of unit broadening, K/V entropy reduction, and residual spatial contrasts. However, the DeiT-S Q term changes point-estimate sign for B11−B0 and its intervals cross zero. On an independent 400-image cohort and four checkpoints, three prespecified directions persist while H4 head-ranking increments over activation × gradient are negative throughout. Continuous component-replacement correlations are positive, but the original all-family claim is not supported; stratified A/V and residual-type diagnostics are reported separately. Reference-aware Taylor better ranks the prescribed region-induced replacement of designated head activations. Dual indexing increases structural resolution and auditability without automatically improving finite-intervention prediction.

Keywords: Vision Transformer; Neural Network Scanner; Gradient × Input; local Jacobian; structural audit

1. Introduction

Vision Transformers (ViTs) divide images into patch tokens and build representations through self-attention and feed-forward layers (Dosovitskiy et al., 2021). This architecture models relationships across positions while making internal computation difficult to interpret. Token-channel units represent responses at particular positions, attention heads transmit information across positions, and residual connections combine inputs with branch outputs. Understanding their relationships calls for a representation that preserves input locations and applies to different internal objects.

Existing visualizations commonly show feature maps, attention weights, or class-related regions. Each view serves a purpose, but their resolutions, targets, and propagation rules differ. A forward feature map shows a channel's response on the token grid, whereas attention weights describe allocation between tokens. Neither is the same object as an internal scalar's dependence on the original image. Mapping internal targets to common input coordinates provides a shared basis for comparing modules. Here, a "path map" is a local differential term for a specified tensor use, not a unique causal path.

Neural Network Scanner (NNS) associates a neuron with a learning image that preserves input positions (Dou et al., 2023). With exact local derivatives, its layerwise scale propagation is an application of the Jacobian chain rule; for a scalar target, the endpoint learning image is Input × Gradient. This identity fixes the local measurement primitive. We retain two free indices beyond that total map. Input coordinate i records where local dependence occurs, while concrete consumer-use edge e records the specific use through which the term passes. Summing out e prevents a total map from distinguishing the spatial allocation of Q/K/V or identity/branch terms; summing out i prevents an edge scalar from recovering component entropy, spatial TV, or signed spatial cancellation. The target registry fixes condition s, and a structurally valid cut binds the terms back to the same total local differential. The dual-indexed family therefore measures redistribution over specified uses and input positions and audits the declared cut contract.

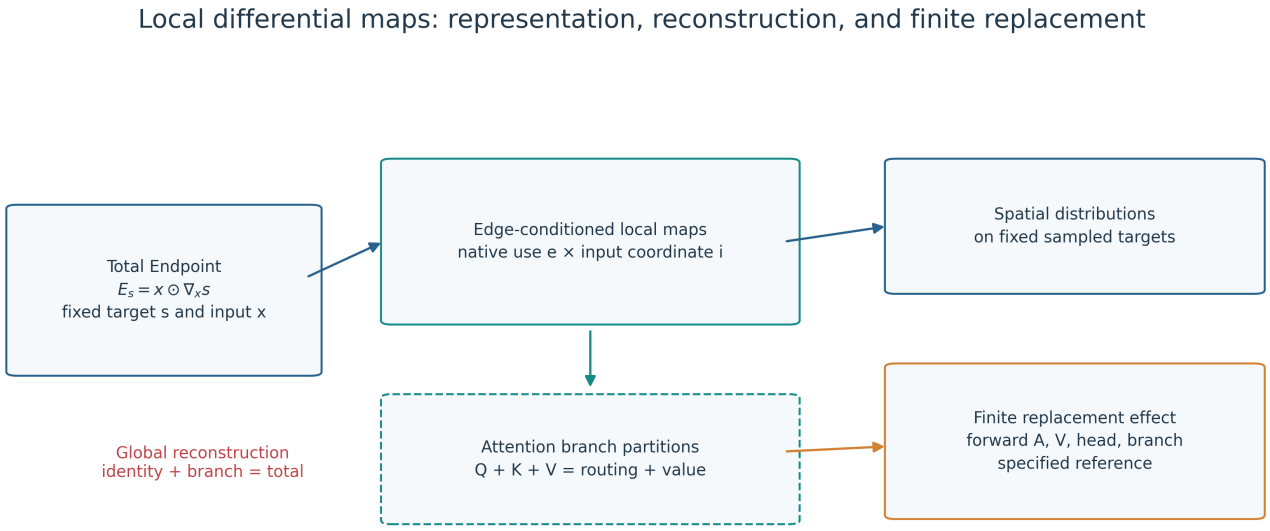


Figure 1. Left: Total Endpoint for one fixed target and input point. Middle: retain native-use identity and input coordinates, with a separate definition for differential partitions inside attention. Right: compare spatial distributions and prescribed finite replacement effects. Global closure includes the residual bypass; a branch partition reconstructs only the attention branch contribution.

The overarching question is how NNS can represent the local input dependence of an internal ViT target as computable, decomposable, and verifiable structural-path maps, and how these maps vary with depth and rank candidates in prescribed finite-replacement tasks. We address four questions:

1. **RQ1: What is the mathematical object represented by an NNS learning image?** We relate layerwise scale propagation to the Jacobian chain rule and Input × Gradient, including the effects of non-homogeneous operations, fixed tokens, and the input-coordinate origin.
2. **RQ2: How can NNS be extended to a structured ViT computational graph?** We define internal scalar targets, actual graph edges, complete computational cuts, and additive local differentials for attention, multi-head structure, and residual paths.
3. **RQ3: How do structural paths vary with depth?** In frozen target samples from two fixed pretrained checkpoints, we analyze spatial broadening, routing/value, and identity/branch trends and compare trained and random parameter states together with target and sampling controls.
4. **RQ4: To what extent do clean-only local-map scores proxy prescribed finite-replacement effects, and how does ranking change when actual activation displacement is included?** H4 increments over

activation × gradient are negative at all four checkpoints; greater structural resolution does not automatically yield predictive gains.

The method contribution has two parts. First, a target-conditioned dual-index local differential representation specifies distinct reconstruction contracts for global cuts and branch partitions. Second, a structural audit checks ViT use identity, omissions, and duplication and compares component maps with an independent analytic reference. The empirical analysis separately addresses spatial distributions in fixed checkpoints. Total maps describe overall broadening; paired maps describe attention and residual spatial differences. Path maps do not outperform activation × gradient in H4, so increased structural resolution does not imply improved intervention prediction.

We next delimit the related methods and our positioning, derive the general Jacobian representation and ViT structural decomposition, and organize experiments around identity and closure, depth trends, structural paths, and finite intervention.

2. Related Work

2.1 NNS and learning images

Neuron visualization examines internal responses through input patterns or observed samples. NNS organizes a neuron's computation as an input-coordinate learning image and represents convolutional channels through activation-weighted aggregation (Dou et al., 2023). This organization maps internal objects with different native spatial structures into common image coordinates. We retain "learning image" as the historical term and use "local input-dependence map" in the mathematical development to emphasize a first-order response at the observed input.

2.2 Local gradients, path integration, FullGrad, and hidden-unit attribution

Gradient × Input is the input-weighted gradient at the observed point (Shrikumar et al., 2017). Integrated Gradients averages gradients along a reference-to-input path and satisfies its corresponding completeness axioms (Sundararajan et al., 2017), while conductance extends path attribution to hidden units (Dhamdhere et al., 2018). FullGrad combines input-gradient and layerwise bias-gradient components into a complete representation and then aggregates them into a saliency map (Srinivas and Fleuret, 2019). Endpoint uses the true current-point Jacobian without path integration and does not add bias terms to its input map. Conductance path completeness, the FullGrad function-value representation, and local-differential closure on our valid cuts are distinct objects.

DAVE decomposes a ViT input gradient into a direct effective input-output transformation and a local-variation term, then aggregates the retained transformation over an input distribution (Wróbel et al., 2026). It is a close precedent for ViT-specific gradient decomposition and spatial attribution; it does not retain a family of input-coordinate maps indexed by internal consumer uses.

2.3 Transformer token contribution and relevance propagation

GlobEnc incorporates attention, residual, and normalization components into layerwise token attribution and aggregates them across encoder layers (Modarressi et al., 2022). ALTI decomposes token interactions over a whole attention

block, including the residual and LayerNorm, and aggregates them by rollout; its published formulation does not include the classifier (Ferrando et al., 2022). DecompX propagates token-representation components through nonlinear feed-forward layers and the classification head and recomposes representations and logits under its bias-allocation rules (Modarressi et al., 2023).

AttnLRP derives relevance rules for softmax, bilinear attention, normalization, and latent attribution (Achtibat et al., 2024). LibraGrad prunes and rescales backward paths under a FullGrad-completeness objective for ViT attribution (Mehri et al., 2025). HiLRP reduces diverse ViT attention and resolution-changing modules to four conserving primitives and verifies conservation to machine precision, while GradSkip explicitly allocates relevance between attention and residual paths (Nishankar et al., 2026; Buratti et al., 2026). Unpack produces component interactions, named end-to-end paths with K/Q/V composition labels, and token attribution without gradients (Chen et al., 2026). These works establish direct precedents for Transformer component propagation, Q/K/V-aware analysis, skip-aware allocation, and exact conservation. Our comparison therefore concerns the retained indices and closure object: the model's unmodified local derivatives, input-coordinate terms for concrete consumer uses, and additive reconstruction of the same local differential on a valid cut.

2.4 Activation patching and edge attribution

Activation patching replaces actual internal variables with clean/corrupted reference states and measures finite output changes; its result depends on the corruption, patch location, and metric (Zhang and Nanda, 2023). Attribution patching approximates a component patching effect with the inner product of activation displacement and a local gradient, and Edge attribution patching applies this approximation to computational edges (Syed et al., 2024). AtP* further treats approximation errors caused by attention saturation and cancellation (Kramár et al., 2024). These methods directly evaluate or approximate reference-specified finite effects. We report clean-only path maps, reference-aware Taylor quantities, and actual replacement effects separately and do not substitute structural closure for finite-intervention utility.

Edge-based analysis and internal intervention already extend to vision. Vi-CD discovers and validates sparse residual-stream edge circuits in ViTs and OpenCLIP by activation patching (Żukowska et al., 2026), while CAAP constructs ViT patch attributions by inserting source-image activations into a neutral context (Izadi et al., 2026). Actual edges and ViT activation patching are established precedents.

2.5 Visual attribution and position of this work

Grad-CAM provides class-conditioned localization through gradient-weighted feature maps (Selvaraju et al., 2017), attention rollout combines attention flow across layers (Abnar and Zuidema, 2020), and Chefer et al. (2021) design relevance propagation for Transformers. These approaches, Gradient × Input, token decomposition, LRP, and finite replacement address different explanation objects. Table 1 focuses on five nearest method families; the complete matrix is retained in the supplement. "Not reported" refers only to the versions reviewed. Live checks were performed on 23 September 2026 (PRC); DAVE, Vi-CD, CAAP, HiLRP, GradSkip, and Unpack are identified by their corresponding arXiv preprint versions.

Table 1. Object-level comparison with five nearest method families

Method family	Primary object	Index retained	Reconstruction target	Empirical role
Conductance	Reference-to-input path attribution through hidden units	Hidden unit and input	Output difference across a layer	Hidden-unit attribution
DecompX / LRP	Token representation or relevance components	Token/operator, method-dependent	Representation, logit, or conserved relevance	Token attribution and localization
EAP / AtP* / Vi-CD	Reference-specific component or edge effects	Component/edge	Patching effect or circuit fidelity	Circuit localization and intervention
DAVE	Distribution-aware ViT input-gradient decomposition	Input position and gradient term	Effective input–output map	ViT spatial attribution
Unpack	Component interactions and named K/Q/V paths	Component/path and token	Method-specific credit decomposition	Composition and token credit
NNS–ViT	Current-point local differential by concrete use and input position	Consumer use e and input coordinate i	Branch contribution or same-target total differential, according to the declared cut	Structural audit and spatial comparison

Conductance uses output difference for path completeness; DecompX/LRP reconstruct representations or relevance; EAP/Vi-CD address reference-specific edge effects; DAVE decomposes ViT input gradients; Unpack retains named component paths. The technical difference studied here is the joint representation of current-point differentials by concrete use and input position, together with implementation-consistency checks for a declared cut contract. Gradient × Input, component decomposition, and activation patching are established methods.

3. Jacobian Interpretation of NNS

3.1 General computational-graph representation of NNS

Let the network be a directed acyclic computational graph $G = (V, E)$, with input x and scalar target $s(x)$. A neuron produces a scalar value, whereas its learning image retains the input coordinates on which that scalar depends. Figure 1 first separates total Gradient × Input from our dual-indexed expansion; coordinate initialization and original NNS layerwise propagation diagrams remain in the supplementary figure bank. We denote the final scalar map by $E_s(x)$ and call it the NNS local input map.

Let the image input $x \in \mathbb{R}^d$ and scalar gradient $\nabla_x s \in \mathbb{R}^d$ be column vectors. For $u(x) \in \mathbb{R}^m$, the learning-image matrix is $\mathcal{L}(u; x) \in \mathbb{R}^{m\times d}$. Initialization and propagation are

$$\mathcal{L}(x; x) = \operatorname{diag}(x), \qquad \mathcal{L}(v; x) = \sum_a J_{f,u_a}(x)\mathcal{L}(u_a; x), \quad v = f(u_1, \dots, u_r). \tag{1}$$

Proposition 1 (scalar-target equivalence). If all local Jacobians are evaluated at the same current forward state and propagation introduces no normalization, truncation, or relevance redistribution, then

$$\mathcal{L}(u; x) = J_{u,x}(x)\operatorname{diag}(x), \qquad \mathcal{L}(s; x) = \nabla_x s(x)^\top \operatorname{diag}(x), \qquad E_s(x) = \mathcal{L}(s; x)^\top = \operatorname{diag}(x)\nabla_x s(x). \tag{2}$$

Proof. At the input, $J_{x,x} = I$. In topological order, substitute each predecessor's expression into the propagation rule and apply the multi-input chain rule. Transposing the scalar node's learning-image row gives the column-vector Endpoint. For linear, convolutional, and current-ReLU-gate operations, this recovers the original NNS coefficients.

MaxPool uses the current forward winner. At ReLU zero and MaxPool ties, we use the framework's fixed derivative convention; smooth regions use ordinary derivatives.

Endpoint describes the target's local first-order dependence on coordinate scaling: changing x_i to $(1+\epsilon)x_i$ produces first-order response $\epsilon E_{s,i}$. In general,

$$\sum_i E_{s,i} = x^\top \nabla_x s(x) \neq s(x). \tag{3}$$

Euler's theorem gives equality for differentiable degree-one homogeneous functions. A general ViT containing biases, fixed tokens, positional embeddings, LayerNorm, softmax, and GELU does not satisfy this condition. Endpoint reconstructs local input-Jacobian decomposition rather than all sources of the target or a complete function-value decomposition. The CLS token, positional embeddings, and patch/LN/QKV/MLP/classifier biases have no separate image-coordinate terms, but they change the current function and Jacobian.

If $x = (r-\mu)/\sigma$, then

$$x_i \frac{\partial s}{\partial x_i} = (r_i - \mu_i)\frac{\partial s}{\partial r_i}. \tag{4}$$

Endpoint therefore requires no externally selected intervention reference, while depending on the model's input coordinates and normalization origin.

4. Dual-Indexed Local Differentials for Internal ViT Targets

4.1 Target registry and the dual-indexed object

The target registry fixes a scalar readout s, graph location, and aggregation rule for each key. Registered objects include token-channel units, channel aggregates, attention heads and Q/K/V terms, routing/value terms, residual identity/branch terms, MLP branches, and class logits. The registry specifies comparable target semantics; it does not enumerate every neuron. For patch activations in channel c, an aggregate target can be defined as

$$s_{\text{energy}} = \frac{1}{2}\sum_{t=1}^{196} h_{t,c}^2, \qquad \nabla_x s_{\text{energy}} = \sum_t h_{t,c}\nabla_x h_{t,c}, \qquad E_{\text{energy}} = \sum_t h_{t,c} E_{h_{t,c}}. \tag{5}$$

This is the Jacobian form of original NNS activation-weighted filter aggregation; energy is an aggregate scalar rather than one neuron. Targets also include patch sums, internal head coordinates, final CLS channels, ground-truth logits, and frozen clean-top1 logits. Equal channel indices across models do not imply semantic correspondence.

The registry specifies target location and readout semantics; registration alone does not make every cut applicable. Auditing the module, location, and kind fields of the 457/385 registered targets gives the following rules.

Target type or position	Admissible global cut	Admissible branch partition	Inapplicable condition
Final class logit	Identity plus a complete branch partition at a specified residual	Attention branch upstream of that residual	None
Internal scalar after a residual output	An upstream residual cut intercepting every input path	Corresponding upstream branch	Cut downstream of the target

Target type or position	Admissible global cut	Admissible branch partition	Inapplicable condition
Target before a residual or inside a branch	A cut determined by its ancestor subgraph	Only when the target is downstream of that branch	A later residual equality cannot be applied
Energy or patch-sum aggregate	Determined by the actual module/location readout	Determined by actual ancestry	Registry membership does not grant every cut

Define a concrete use edge as

$$e = (p_e, c_e, q_e, \nu_e), \tag{6}$$

where p_e is the producer, c_e the consumer, q_e the consumer input slot, and ν_e the invocation. Let $z_e(x)$ be the producer output as it enters that consumer use. Let $\bar{z}_e^{(s)}$ be the target cotangent at that input slot before it is merged with other fan-out uses of the same producer. We define

$$\boxed{\mathcal{L}_{e,i}(s;x) = x_i \left[J_{z_e,x}(x)^{\mathsf{T}} \bar{z}_e^{(s)} \right]_i .} \tag{7}$$

Here i and e are free indices and s is the target condition selected by the registry. The Jacobian, cotangent, and z_e are evaluated at the same observed input x and forward state. Fixing e yields one input-position-preserving map; fixing i compares local terms from one input coordinate across concrete uses; summing over a global cut returns the total. This representation retains concrete uses and input positions. Total Endpoint remains the established Gradient $\times$ Input primitive.

One producer tensor may enter several consumers, different slots of one consumer, or repeated invocations. Their upstream $J_{z_e,x}$ values may coincide, but $\bar{z}_e^{(s)}$ belongs to a different downstream use. Merging these cotangents by tensor name discards path identity before measurement, so edge identity includes the consumer, slot, and invocation. Figure 2 places this contract beside concrete ViT cuts.

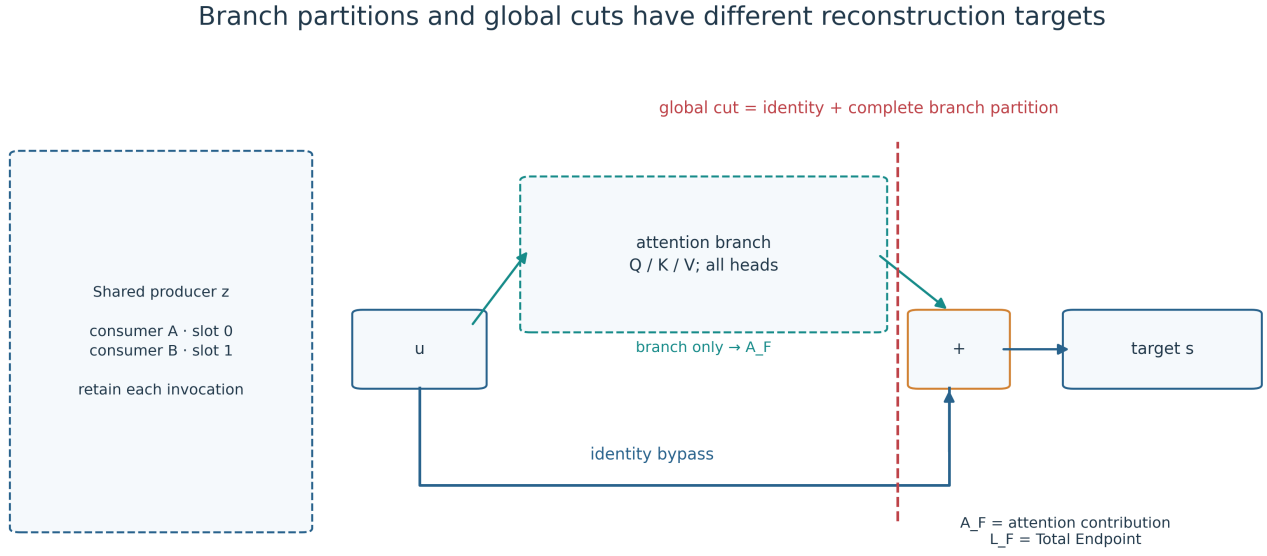


Figure 2. Left: two consumer slots of a shared producer remain distinct. Middle: the lower blue path is the residual identity bypass; the upper path is attention. The green dashed region contains a branch partition and reconstructs

attention branch contribution A_F. The red dashed global cut includes identity and the complete branch partition and reconstructs Total Endpoint L_F. Right: the two reconstruction targets.

4.2 Attention branch differentials and residual global cuts

Figure 3 describes differential partitions inside the attention branch. Figure 4 shows the identity and branch uses at a residual addition.

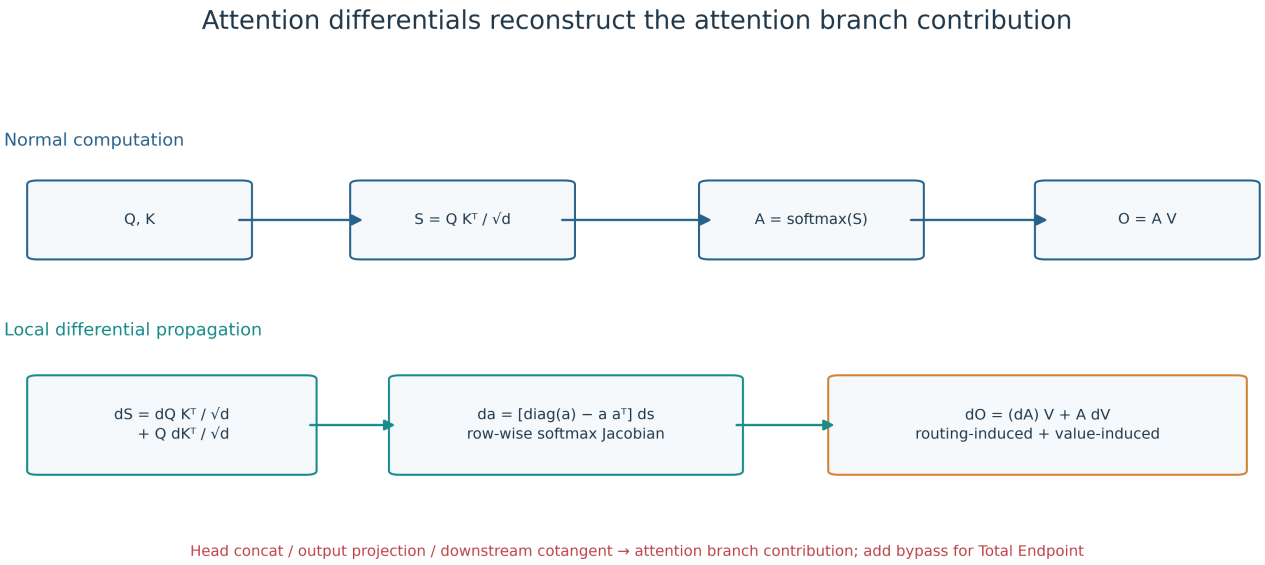


Figure 3. The upper row shows one head's forward computation, with attention probability A and value projection V. The lower row decomposes $\mathrm{d}O_h = (\mathrm{d}A)V + A\mathrm{d}V$ at the same forward point. Q-induced and K-induced terms form the routing-induced differential; the V-induced term is the value-induced differential. After contraction with the downstream cotangent through head concatenation and output projection, these terms reconstruct the attention branch contribution. The global target additionally requires the residual bypass.

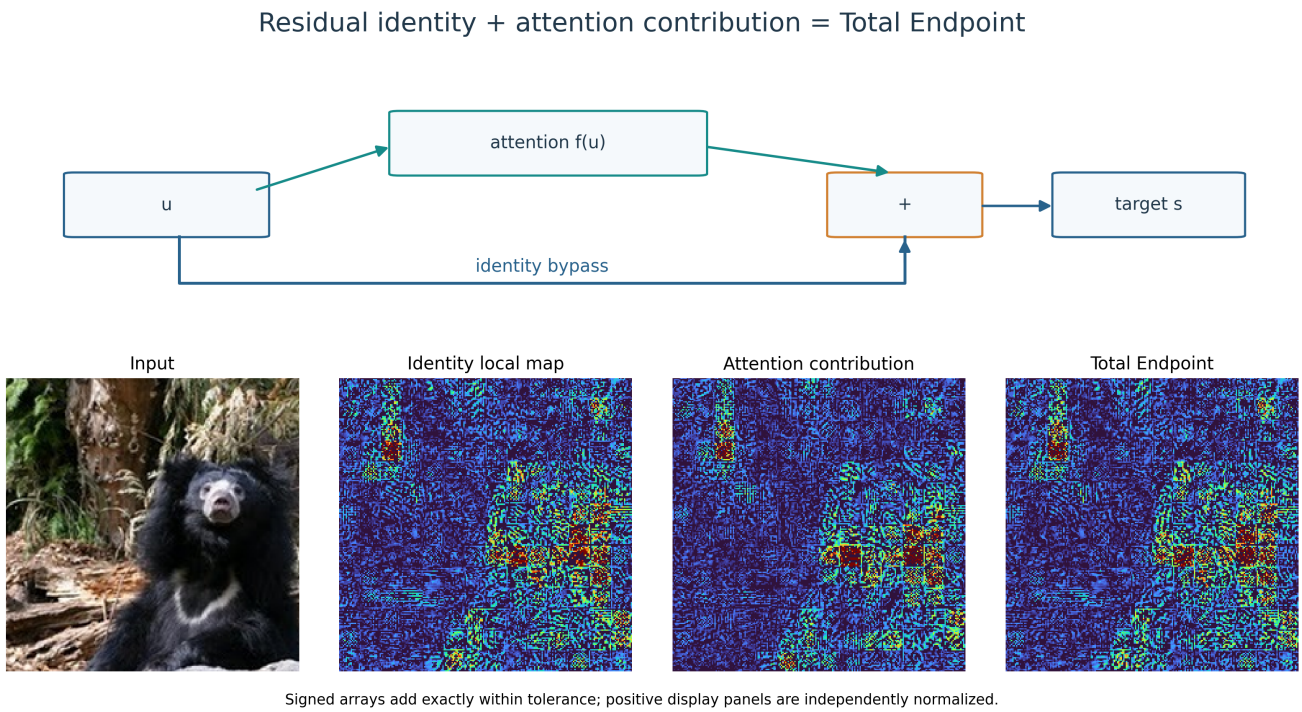


Figure 4. The identity and branch inputs of a residual addition define separate local maps for the same final class target and the cotangent at the addition output. Complete signed maps sum to Total Endpoint. Independently normalized blue positive displays show spatial structure.

For each head, $Q = XW_Q + b_Q$, $K = XW_K + b_K$, $V = XW_V + b_V$, and

$$A = \mathrm{softmax}(QK^\top/\sqrt{d_h}), \quad O_h = AV, \quad \mathrm{d}O_h = (\mathrm{d}A)V + A\mathrm{d}V. \tag{8}$$

Here $\mathrm{d}A = J_{\mathrm{softmax}}(S)\mathrm{d}S$ and $\mathrm{d}S = (\mathrm{d}QK^\top + Q\mathrm{d}K^\top)/\sqrt{d_h}$. Let $a = f(u)$ be the attention branch output, $y = u + a$ the residual output, and $g_y^{(s)}$ the cotangent of final target s at y. Define

$$\mathcal{B}_i^{\mathrm{attn}}(s;x) = x_i \left[J_{a,x}(x)^{\mathsf{T}} g_y^{(s)}\right]_i, \qquad \mathcal{B}_i^{\mathrm{attn}} = \mathcal{B}_i^Q + \mathcal{B}_i^K + \mathcal{B}_i^V = \mathcal{B}_i^{\mathrm{routing}} + \mathcal{B}_i^{\mathrm{value}}. \tag{9}$$

Q/K/V projection slots are native computational uses. The Q-induced, K-induced, and V-induced terms are derivative contributions induced by those slots. The routing-induced differential analytically groups the Q-induced and K-induced terms; the value-induced differential is $A\mathrm{d}V$. These branch-internal differential partitions reconstruct the attention branch contribution and do not alone equal the full class-target G×I. Each head-map cotangent includes concatenation, output projection W_O, and subsequent derivatives to the final target. Components may overlap in input space.

For the residual identity, define

$$\mathcal{L}_i^{\mathrm{identity}} = x_i[J_{u,x}^{\mathsf{T}} g_y^{(s)}]_i, \qquad E_{s,i} = x_i \frac{\partial s}{\partial x_i} = \mathcal{L}_i^{\mathrm{identity}} + \mathcal{B}_i^{\mathrm{attn}}. \tag{10}$$

The identity term uses the cotangent at the addition output. Substituting $\nabla_u s$, which already includes the branch, duplicates paths. At the second residual addition, carry plus MLP branch reconstructs the same Total Endpoint. The tables distinguish the mathematical objects and implemented reconstruction targets.

Object class	Examples	Native forward tensor/use	Direct replacement	Reconstruction or estimand
Native computational edge	Residual add identity/branch input; head-to-concat input; Q/K/V projection slot	Yes	Only under a specified runtime contract	Branch or full-target differential, depending on the edge set
Analytic differential partition	Q-induced, K-induced, routing-induced, value-induced terms	No; groups of derivative terms	No	Attention branch contribution
Intervenable forward variable	Attention probability A; value projection V; head output; residual branch activation	Yes	Yes	Prescribed finite replacement effect
Aggregate or ranking score	Map absolute mass; regional map score; reference-aware Taylor score	No	Not applicable	Specified statistical estimand

Named component set	Required terms	Reconstruction target
heads_branch	All head maps	Attention branch contribution $\mathcal{B}^{\text{attn}}$ (A_F in code)
qkv_branch	Q-induced + K-induced + V-induced	Attention branch contribution $\mathcal{B}^{\text{attn}}$ (A_F)
routing_value_branch	routing-induced + value-induced	Attention branch contribution $\mathcal{B}^{\text{attn}}$ (A_F)
heads_with_bypass / qkv_with_bypass / routing_value_with_bypass	Identity + the corresponding complete branch partition	Total Endpoint E_s (L_F in code)
residual1 / residual2	Identity + attention / carry + MLP	Total Endpoint E_s (L_F)

4.3 Structurally valid cuts and pointwise differential closure

For a specified tensor-use DAG, input node x, and scalar target s, let C be a set of concrete use edges. The set is a **structurally valid cut** for that graph and target if every directed use-path from x to s in the target's ancestor subgraph crosses exactly one edge in C. This property depends on graph structure, target location, and use identity, not on the current input value or whether a derivative happens to vanish at that point.

At fixed input, parameters, and derivative convention, C has **pointwise differential closure** for the target when

$$\nabla_x s = \sum_{e \in C} J_{z_e, x}^{\mathsf{T}} \bar{z}_e^{(s)}, \tag{11}$$

Numerical closure tests implementation consistency at that point. It cannot detect an omitted structural path whose derivative vanishes at the tested input.

Proposition 2 (a structurally valid cut is sufficient for pointwise differential closure). The scalar target must lie downstream of the cut, and every structural input-to-target path must pass through the declared cut. If the graph is differentiable at x under a fixed derivative convention, all components use the same forward state, and C is structurally valid in the target's ancestor subgraph, then, for every input coordinate i,

$$\boxed{\sum_{e \in C^{\text{global}}} \mathcal{L}_{e,i}(s; x) = x_i \frac{\partial s(x)}{\partial x_i}.} \tag{12}$$

Inputs to a multi-input operator are counted separately; repeated uses of a shared tensor are distinguished by the tuple e rather than tensor name. The proposition does not apply to a target downstream of, outside, or reachable around the claimed cut.

Proof. Treat the tensors at the cut uses as components of one multivariate map $z_C(x) = (z_e(x))_{e\in C}$ while retaining a use-specific cotangent at each consumer input slot. Structural validity implies that the target ancestor subgraph factors as $s = \tilde{s}(z_C(x))$: no input path bypasses C, and no path is counted twice in C. The multivariate chain rule gives

$$\nabla_x s = J_{z_C,x}^{\mathsf{T}} \nabla_{z_C} \tilde{s} = \sum_{e\in C} J_{z_e,x}^{\mathsf{T}} \bar{z}_e^{(s)}. \tag{13}$$

Here $\bar{z}_e^{(s)}$ is assigned to that concrete use; different consumers, slots, or invocations of a shared producer are not merged first. Multiplication by x_i gives the Endpoint equality. A coordinate expansion supplies an intuitive path view, but the proof does not enumerate tensor-computation paths. Cutting only $f(u)$ omits the identity bypass in $y = u + f(u)$; using $\nabla_u s$, which already includes the branch, as the identity term and then adding the branch duplicates it.

A minimal counterexample is $s(x) = x + x^2$. At $x = 0$, omitting the square branch leaves derivative $1 = \mathrm{d}s/\mathrm{d}x$, so the pointwise numerical closure error is zero. The structural path through the square branch still exists, and the edge set that omits it is not a structurally valid cut. Pointwise equality therefore cannot certify structural completeness.

Method ablation table: index necessity and object degeneration

Retained structure	Observable	Lost when removed	Degeneration
$[e, i]$ with registered s and valid C	Path-specific spatial maps that reconstruct one total local differential	Nothing in the declared object	Auditable dual-indexed map family
no i	Per-edge aggregate magnitude or score	Spatial allocation and signed spatial cancellation	Edge scalar
no e	Total input-coordinate dependence	Routing/value, head, and identity/branch identity	Total Gradient × Input
no registered s	Maps from ad hoc scalar readouts	Comparable target semantics and reproducible atlas coverage	Unregistered target collection
no valid-cut contract	Named component maps	Detection of omitted or duplicated live paths	Unverified component list

This is a structural rather than empirical ablation. Removing i hides spatial broadening and within-path redistribution; removing e adds different paths into one total map; removing the registry for s removes fixed cross-target semantics; and removing the valid-cut contract leaves visually plausible maps without evidence against omitted or duplicated paths.

Structural audit coverage

Audit layer	Reconstruction / question	Direct evidence	Scope
Formula identity	Same-target Total Endpoint	5,000 numerical comparisons; 88 operator directions	Declared exact local-derivative rules
Branch closure	Attention branch contribution A_F	28,800 branch-partition closures	Fixed tested attention graphs
Global closure	Total Endpoint L_F	48,000 bypass-inclusive or residual closures	Fixed tested global cuts
Existing mutations	Omitted head/routing, duplicate identity	9,600 of each error, all rejected	Three prescribed fault classes
Independent analytic reference	Q/K/V, routing/value, heads, identity, branch, full	32 component comparisons and 9 central-difference directions	Three tiny attention/residual graphs; float64
Use-identity mutations	Wrong slot and merged fan-out	Four mutations rejected after correct controls pass	Shared-producer toy and Q/K slot exchange
Numerical directions and summaries	Local correctness	6,144 analytic JVP; 384 attention-detach consistency checks; 230,400 summary fields	Directional/summary checks, not fully independent component construction

The analytic implementation is independent of the main scanner; a separate driver compares the actual scanner component by component. Nonzero asymmetric fixtures distinguish wrong slots and prematurely merged uses. Correctness is limited to the tested graphs and fault types. Arrays, per-component errors, and mutation records appear in Supplement S17.

4.4 Computational-graph dependence and non-uniqueness

Closure in Proposition 2 is relative to a specified computational graph, cut, and internal coordinate system. An equivalent reparameterization can preserve the end-to-end function and total Endpoint while changing coordinate-wise or path-wise terms. Let $x = (x_1, x_2)^\top$, hidden state $h = x$, and output $s = (1, 1)h = x_1 + x_2$. A cut over the two hidden coordinates gives

$$E_1 = (x_1, 0)^\top, \qquad E_2 = (0, x_2)^\top. \tag{14}$$

Now use the invertible transform $R = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and rewrite the same function with $h' = Rh$ and $w' = R^{-\top}(1, 1)^\top = (1, 0)^\top$, so $s = w'^\top h'$. The coordinate-path maps become

$$E'_1 = (x_1, x_2)^\top, \qquad E'_2 = (0, 0)^\top. \tag{15}$$

Both implementations satisfy $E_1 + E_2 = E'_1 + E'_2 = (x_1, x_2)^\top$, but their terms differ. Changing a valid cut likewise changes the named path groups. Our Q/K/V, head, and residual maps are therefore decompositions of specified actual edges in the frozen ViT implementation, not unique explanations determined solely by the end-to-end function, and they are not generally implementation invariant. The tested cuts and mutation classes check omissions and duplication in that representation. Their terms need not be unique across equivalent graphs.

4.5 Mass, cancellation, and the Value-only approximation

On the raw normalized-input RGB tensor, define the primary spatial measure at pixel p as

$$M(p) = \sum_{c \in \{R,G,B\}} |E_{p,c}|, \tag{16}$$

and retain $M^+(p) = \sum_c \max(E_{p,c}, 0)$ and $M^-(p) = \sum_c \max(-E_{p,c}, 0)$ for sign analysis. Patch mass is $m_j = \sum_{p \in P_j} M(p)$ and, when total mass is nonzero, $p_j = m_j / \sum_k m_k$. Zero-mass cases propagate as NA rather than being regularized by an ε-defined distribution. Normalized spatial entropy is $H = -\sum_j p_j \log p_j / \log 196$. For patch tokens, own-token locality is the mass in the Chebyshev-radius-one neighborhood around the target patch minus the corresponding uniform-area baseline; it is undefined for CLS. Normalized distance averages Euclidean distances from the target patch center after division by $13\sqrt{2}$.

For the complete RGB tensor, also define $m(E) = \sum_i |E_i|$ and $a(E) = \sum_i E_i$. Within-map and between-path cancellation are

$$C^{\mathrm{map}} = 1 - \frac{|\sum_i E_i|}{\sum_i |E_i|}, \qquad C^{\mathrm{path}} = 1 - \frac{\|E^{(1)} + E^{(2)}\|_1}{\|E^{(1)}\|_1 + \|E^{(2)}\|_1}. \tag{17}$$

Routing scale is $\|E^{\mathrm{routing}}\|_1/(\|E^{\mathrm{routing}}\|_1 + \|E^{\mathrm{value}}\|_1)$, and identity scale is defined analogously. Alongside these ratios, we report absolute mass for routing/value/full and identity/branch/output. The Value-only approximation retains $A\mathrm{d}V$ and omits $(\mathrm{d}A)V$; its Endpoint relative error is $\|E^{\mathrm{full}} - E^{V}\|_2/\|E^{\mathrm{full}}\|_2$. JVP discrepancy uses the corresponding complete directional response as denominator over frozen directions. An error near 0.95 indicates a discrepancy on the scale of the full response.

These scales are absolute-mass ratios of current-point local differentials, not causal contribution fractions or functional weights. Own-token locality uses Chebyshev radius 1, the one-hop neighborhood on the 14×14 patch grid; the post-hoc radius-0/1/2 sensitivity analysis checks only this descriptive choice.

Statistics use complete signed RGB tensors; zero denominators or norms yield NA. The blue display convention is described in the experimental settings.

4.6 Finite replacement and region-induced replacement of designated head activations

Define the finite component effect as $e_k = s(x) - s(x; z_k \leftarrow z_k^{\mathrm{ref}})$. Table 6 compares m_k with $|e_k|$ by within-image Spearman correlation. Intervenable variables are attention probability A, value projection V, head output, and residual branch activation. Routing-induced and value-induced differentials supply the corresponding local maps; derivative terms are not replacement tensors.

We use five names consistently. Total Endpoint is the full-target map E_s. An edge-conditioned local map fixes one use in $\mathcal{L}_{e,:}$. A branch contribution map sums a complete branch partition. A regional map score aggregates a specified map within a candidate region. A reference-aware Taylor score combines the clean cotangent with the actual activation displacement. Tables 6–10 use the following formulas, where k denotes a native use or declared differential group and x_P uses reference pixels only within region P.

$$E_s = x \odot \nabla_x s, \qquad m_k = \sum_i |\mathcal{L}_{k,i}|, \qquad R_k(P) = \sum_{i \in P} |\mathcal{L}_{k,i}|, \qquad R_{\mathrm{total}}(P) = \sum_{i \in P} |E_{s,i}|, \tag{18}$$

$$R_{\nabla_x}(P) = \sum_{i \in P} |\partial_i s|, \qquad T_k = \left|\langle g_k, z_k(x) - z_k^{\mathrm{ref}} \rangle\right|, \qquad T_k(P) = \left|\langle g_k, z_k(x) - z_k(x_P) \rangle\right|. \tag{19}$$

Table	Method or quantity	Formula and information
6	Component-map association	Spearman$(m_k, \lvert e_k \rvert)$ over the fixed component family within each image; clean-only map mass
7	Local-map / Taylor head rank	Rank m_k / T_k; Capture@K is top-K absolute effect divided by the candidate top-K effect; Regret@1 is the best minus selected effect
8	Paired regional AUC difference	AUC ranked by R_k minus AUC ranked by R_{total}, R_{∇_x}, $T_k(P)$, or random order
9	Absolute isolated-effect AUC	Trapezoid over replaced area 0, 1/16, 2/16, 4/16; divide by 0.25; average heads within image
10	Candidate regret / logit ratio	$1 - \mathrm{AUC}/\mathrm{AUC}_{\mathrm{oracle}}$ / AUC/$\lvert s(x) \rvert$; zero denominator is NA

The normalized-coordinate gradient baseline uses $\nabla_x s$ and is called the normalized-coordinate gradient baseline here. The raw-pixel gradient $\nabla_r s = \nabla_x s/\sigma$ is evaluated separately in coordinate sensitivity. Regional tasks fix three heads in block 5, a shared 4×4 candidate grid, and budgets of 1, 2, and 4 regions. Block 5 and each model's first, middle, and last head indices were selected before outcome access to cover positions within that block; they were not selected by effect and do not represent all heads. The mixed input supplies the selected head activation, which is inserted into a clean forward. Other heads and the identity stay clean, and all downstream nodes are recomputed. Random ranking takes absolute effects and integrates within each repeat before averaging 20 AUCs. Clean-only map scores and reference-aware Taylor use different information. The estimand is a two-stage replacement effect for designated head activations, not a total pixel intervention effect.

Complete replacement pseudocode and variable definitions appear in Supplement S16.

5. Experimental Setup and Numerical Validation

5.1 Models and images

Images come from the 50,000-image ImageNet ILSVRC2012 public validation set and are identified by official WNID labels, original filenames, byte counts, and SHA-256 values. After excluding prior exposures, fixed SHA-256 ordering selects one image from each of 528 classes, divided into 64 discovery, 64 calibration, and 400 evaluation images with no overlap. The 100-image unit study is a fixed evaluation subset. Each block includes CLS, 15 stratified patch tokens, and 32 channels, yielding 6,144 units per image. Correctness and confidence do not determine inclusion; misclassified and low-confidence images are retained.

The independent cohort contains one image from each of another 400 classes, disjoint from the old cohorts and concurrent 500-image cohort. Four public checkpoints include the original models and two additional training recipes. They are fixed parameter states, not independent training seeds or replication by an external team. Exact timm names, training sources, sizes, SHA-256 values, overlap matrices, and randomization rules appear in Supplement S15.

Table 2. Fixed-model settings, descriptive scope

Model / setting	ViT-B/16	DeiT-S/16
Blocks / heads / hidden dimension	12 / 12 / 768	12 / 6 / 384
Registered scalar targets/image	457	385
Evaluation images	400	400 (same IDs)
Resize / crop	248 / 224	248 / 224
Normalization mean	[0.5, 0.5, 0.5]	[0.485, 0.456, 0.406]
Normalization std	[0.5, 0.5, 0.5]	[0.229, 0.224, 0.225]

5.2 Visualization settings

Concept and layerwise figures use a sloth-bear image with a recognizable subject; method comparisons also include a mantis and a little blue heron. Images are selected for presentation readability, while population conclusions retain the complete cohorts. To relate learning images to forward features, each layer selects the channel with highest spatial variance across patch activations and uses its energy as the scalar target. Figures report the channel index and actual 14×14 forward feature map; selected channels may differ across layers.

Main learning images use a deep-blue, cyan, yellow and red positive view. With $P = \max(\sum_c E_c, 0)$, displayed values are $\sqrt{\min(P/q_{.99}, 1)}$, where $q_{.99}$ is computed over positive pixels. This transform improves visibility of weak detail while preserving pixel positions and is excluded from statistics. Supplementary figures pair it with signed and absolute views and separately show a patch-max and bilinear-interpolation display variant, distinguishing native fine detail from a smoothed regional view.

5.3 Inferential populations and analysis identities

Images are the primary statistical units. Fixed within-image aggregation precedes 10,000 image bootstraps for pointwise median 95% intervals. The bootstrap does not generate p values or determine the resolution of small p values. One-image-per-class inference is conditional on the fixed class–image sampling rule and does not cover within-class variation. Zero image overlap between E5 and earlier cohorts does not imply zero class overlap. Intervals condition on checkpoints, target sets, references, and replacement contracts, without direct generalization to new neurons, heads, or architectures. Separate sensitivity analyses jointly resample target strata and images. The four-checkpoint check concerns fixed public models. Holm correction does not make the intervals simultaneous.

The original component and regional study jointly adjusts 28 tests. Unit depth has five tests per model, attention four per model, and residuals two per model and type. Four post-hoc residual-TV tests form a separate Holm family. Independent-cohort H1–H4 tests adjust four hypotheses within each checkpoint. Membership, freeze times, and identities appear in Supplement S15. These local plans were frozen before corresponding outcome access, without external preregistration.

Analysis	Unit	Image-level value	Interval	Test	Sidedness	Zero handling	Multiplicity	Identity
Original component association	image	Within-image Spearman$(m_k, \lvert e_k \rvert)$	10,000-replicate percentile interval for the median	Exact sign test	two-sided	$\lvert v \rvert \le 10^{-12}$ is a tie; retained in total n and interval, excluded from the binomial count	Joint 28-test Holm family with regional tests	Primary analysis locally frozen before outcome access
Original regional comparison	image	Paired AUC difference averaged over three fixed heads	same	Exact sign test	two-sided	same	Same 28-test Holm family	Primary analysis locally frozen before outcome access
H1–H4	image	Fixed within-image hypothesis aggregate	same	Exact sign test	two-sided	same	Four-test Holm family within each checkpoint	Prespecified robustness analysis
v7 stratified diagnostics	image	Type-specific, block-specific, or residualized correlation	same	Exact sign test when reported	two-sided	same	Separate post-hoc Holm family	Post-hoc diagnostic

The exact sign test evaluates equal probabilities of positive and negative signs among non-ties. It is a direct test of sign probability and is equivalent to a median-zero test only under the corresponding distributional conditions. Main-text p-value columns report Holm-adjusted values; machine evidence retains full-precision raw and adjusted values. The display $p < 10^{-6}$ is used only when the adjusted value is below that bound.

The 0.10 correlation and 0.05 regional-gain thresholds are researcher-selected operational thresholds. The original composite criterion requires complete attempted coverage, at least 360 valid images, at least 300 nonzero differences,

CI width no greater than 0.30/0.50, a positive estimate meeting the threshold, and original Holm p no greater than 0.05. Regional support requires all four baselines to pass. This differs from requiring the CI lower bound to exceed the threshold. Correlation sensitivity yields 12/11/9 qualifying point estimates at thresholds 0.05/0.10/0.15 versus 12/10/7 lower bounds. None of eight regional gains against Taylor or the corrected absolute-effect random baseline meets 0.025, 0.05, or 0.10. Sensitivity does not alter the original frozen decisions.

Table 3 reports errors and coverage of fixed numerical tests without image-population inference. Original float32 failures and same-coordinate float64 checks are retained.

Table 3. Numerical agreement and structural checks

Compatible network	Max relative L2: float64	Max relative L2: float32	Comparisons
fcn	4.371e-16	1.964e-07	1,000
cnn	4.289e-16	2.189e-07	1,000
cnn_maxpool	4.769e-16	2.427e-07	1,000
residual_mlp	2.686e-16	2.155e-07	1,000
residual_cnn	2.923e-16	1.692e-07	1,000

Structural / directional check	Count	Result
Positive closures	76,800	All pass
Omission/duplication controls	28,800	All rejected
Analytic JVP	6,144	All pass
Operator directions	88	All pass
Atlas float32 finite differences	57,600	36 outside tolerance
Same-coordinate float64 review	36	All pass

5.4 Result order and evidence scope

Results proceed from identity and numerical correctness to normalized-input Endpoint spatial profiles, independent-cohort checks, coordinate sensitivity, and prescribed finite replacement. Closure concerns local differential accounting, spatial distributions describe fixed checkpoints, and replacement effects depend on the specified reference and intervened variable.

6. Depth Patterns of Local Dependence in ViTs

6.1 RQ1–RQ2: NNS equivalence and structural-path closure

The 5,000 original-NNS-compatible comparisons in Table 3 give maximum relative L2 errors of 4.769×10^{-16} in float64 and 2.427×10^{-7} in float32 between layerwise learning-image propagation and same-target Input $\times$ Gradient. Structural validation covers 76,800 positive closure checks, 28,800 omission/duplication controls, 6,144 analytic JVP checks, and 88 operator directions. All positive closures and analytic directional checks pass, all negative controls are rejected, and all 36 float32 finite-difference exceedances pass same-coordinate float64 review. These results establish the numerical identity in Proposition 1 and structural accounting in Proposition 2; Section 7 evaluates finite-intervention utility separately.

This revision's independent analytic reference passes 32 component comparisons and nine central-difference directions, with maximum component absolute error 1.11×10^{-16}. All four new use mutations are rejected; Supplement S17 reports the direct evidence.

6.2 RQ3: Spatial distributions of fixed targets

Figure 5 shows normalized-input Endpoint for different targets on one image. Forward features describe token-grid responses; local maps describe the scalar's current-point response to input-coordinate scaling. Population summaries use fixed sampled targets independently of display selection.

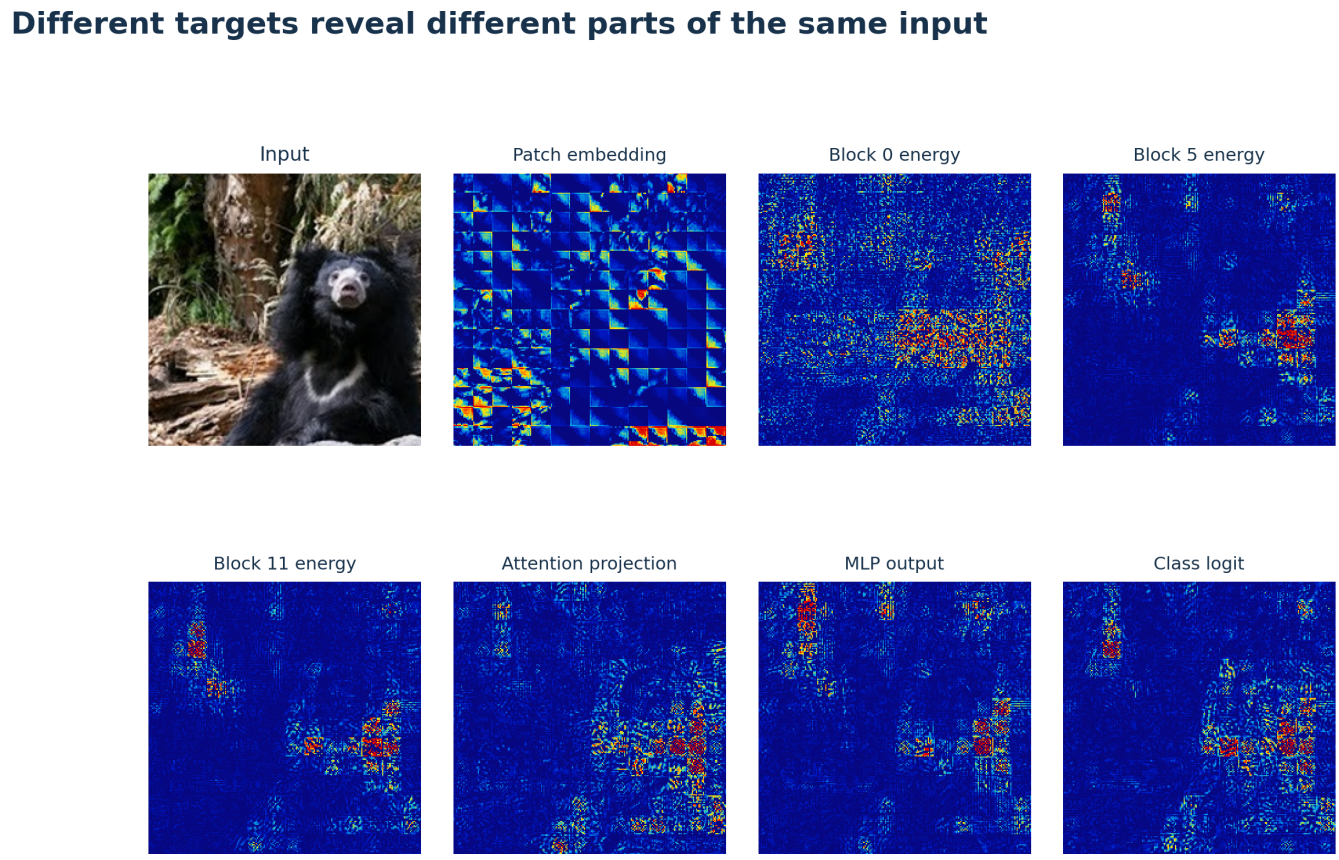


Figure 5. Left-to-right, top-to-bottom: the sloth-bear input and normalized-input Endpoint for patch-embedding energy, block-0/5/11 energy, block-5 attention-projection energy, MLP-output energy, and the ground-truth class logit in DeiT-S/16. Each energy is the half-squared-sum target for the channel with highest forward spatial variance at that site. Map panels show the positive signed RGB sum, clipped at positive P99 and square-root transformed independently. One display image is shown, without intervals or population inference.

Image-level late-minus-early medians and 95% intervals for 100 images per model are listed in ViT-B/16 / DeiT-S/16 order: entropy 0.220 [0.211, 0.227] / 0.225 [0.212, 0.228]; locality -0.356 [-0.363, -0.350] / -0.342 [-0.353, -0.333]; distance 0.131 [0.129, 0.136] / 0.122 [0.120, 0.128]. Each image takes medians over its fixed units in blocks 8–11 and 0–3 before differencing. Locality and distance exclude CLS. These are normalized-input Endpoint distribution profiles, without a causal interpretation of depth.

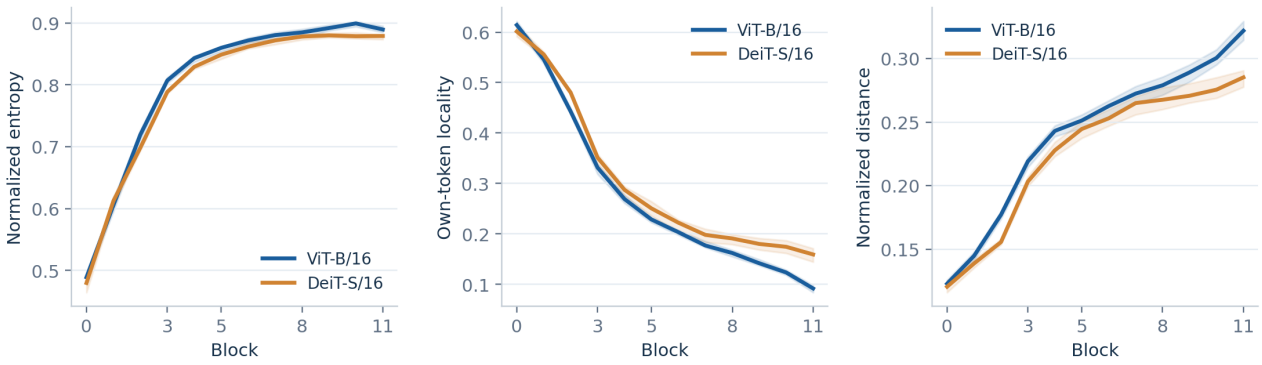


Figure 6. Left, middle, and right panels show normalized-input Endpoint spatial entropy, own-token locality, and normalized distance. At each block, medians over the fixed token-channel targets are computed within images and then summarized by the median over 100 images from the evaluation subset. CLS enters entropy only; locality subtracts the equal-area uniform baseline. Shading shows 95% intervals from 10,000 image bootstraps. Curves correspond to two fixed checkpoints and are descriptive block profiles; late-minus-early tests belong to the five-item family frozen before outcome access.

6.3 RQ3: Joint-index necessity

Total Gradient × Input, edge scalars, and dual-indexed map sets retain different information. Table 4 connects these representations to the observed results. Depth-wise spatial broadening requires only the total map and is therefore a background regularity. Routing and identity shares require edge-level absolute masses and serve as supporting scale results. The spatial differences between full and Value-only attention and between residual output and identity require both path identity e and input position i; they are the core empirical results for the dual-index representation.

Table 4. Representation information and inferential scope

Representation	Retained indices	Recoverable quantity	Evidence / inference scope
Total Endpoint	i	Entropy, locality, distance	100 images/model; fixed sampled targets; image-level inference
Component scalar	use or differential-group label	m_k and relative mass	400 images/model; fixed checkpoint and class target
Dual-indexed map set	label and i	Q/K/V entropy; output–identity TV	400 images/model; QKV prospective robustness; residual TV post-hoc

The alternative-explanation audit uses the same information boundary as the claim. A total map retains i but sums out e, so it cannot recover separate routing/value or identity/branch spatial distributions. An edge scalar retains e but sums out i, so it cannot compute normalized patch distributions, spatial TV, or signed spatial cancellation. Total absolute mass and activation norm can explain overall scale but cannot reconstruct these position-conditional differences. Both Full–Value and output–identity TV compare normalized spatial distributions, so neither is a restatement of total mass. Spatial reorganization therefore requires $[e, i]$; routing share, identity share, and overall broadening do not independently carry the joint-index necessity claim.

6.4 RQ3: Spatial components of the attention branch

Figure 7 separates Q-induced, K-induced, routing-induced, and value-induced differentials. Complete signed terms reconstruct attention branch contribution; positive maps are displays.

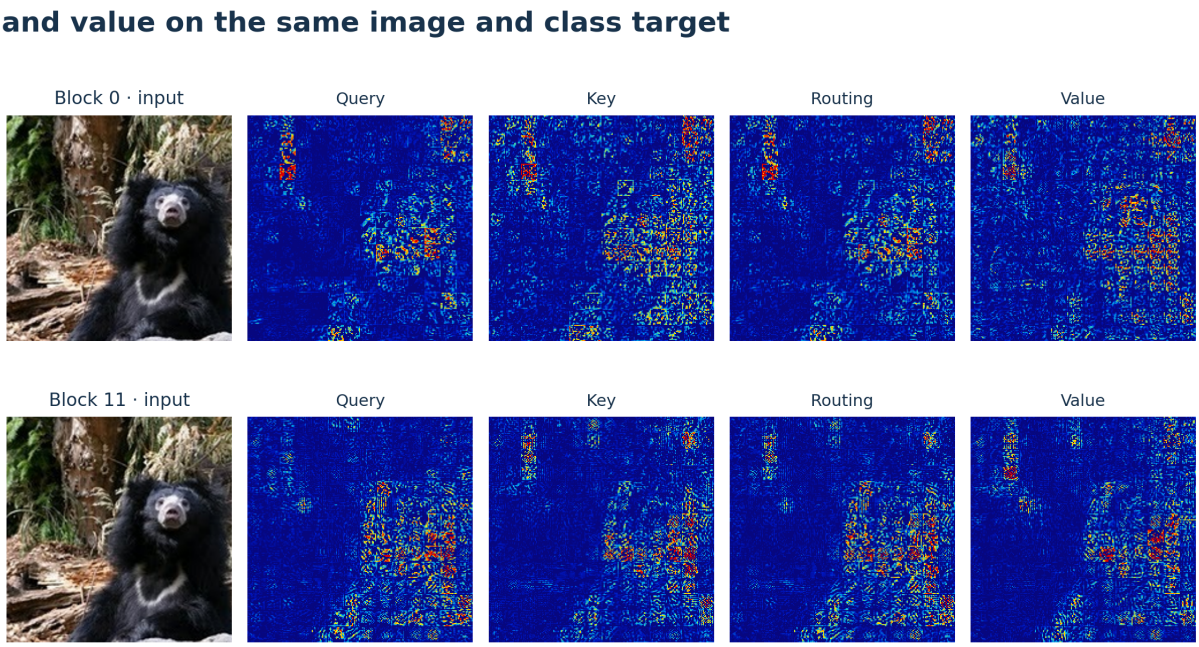


Figure 7. Rows show DeiT-S/16 blocks 0 and 11 for the same sloth-bear input and ground-truth logit. Columns show the input, Q-induced, K-induced, routing-induced, and value-induced maps; routing-induced is the sum of Q and K maps. These normalized-input positive displays are independently clipped at positive P99 and square-root transformed, so color magnitudes are not comparable across panels. This is a single descriptive example without uncertainty bands.

For 400 images per model, routing-relative local differential scale rises with block index, with late-minus-early median differences 0.052 [0.047, 0.059] / 0.135 [0.128, 0.143]. The corresponding increase in Value-only L2 error relative to the full attention branch map is 0.144 [0.133, 0.152] / 0.290 [0.279, 0.301]. These scale measures require component scalars; spatial comparisons additionally require input positions.

The 400-image Q/K/V contrast is B11−B0 and is distinct from four-block phase medians. Normalized-entropy median differences with this revision's descriptive image-bootstrap intervals are vit_b16: Q 0.027 [0.022, 0.031], K -0.059 [-0.064, -0.053], V -0.053 [-0.059, -0.047]; deit_s16: Q -0.004 [-0.008, 0.001], K -0.067 [-0.074, -0.059], V -0.067 [-0.075, -0.061]. K/V decrease in both models; Q remains model-dependent and the DeiT-S interval crosses zero.

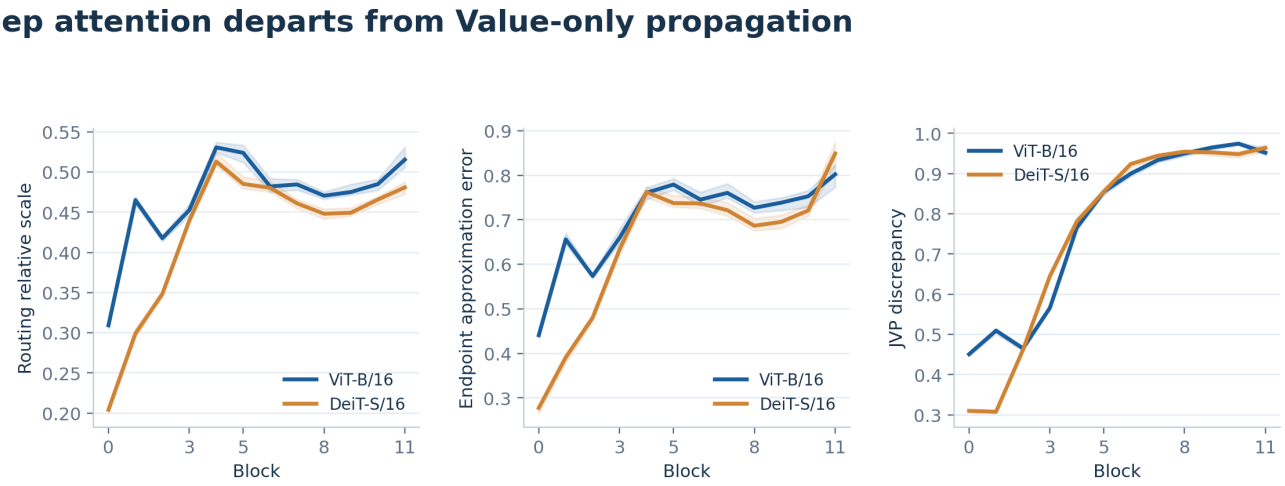


Figure 8. Left panel: routing-induced relative absolute mass. Middle: relative L2 error of the Value-only map against the full attention branch map. Right: JVP discrepancy over frozen directions. Each block takes the within-image median across all heads and then the median across 400 images per checkpoint. Shading shows 95% intervals from 10,000 image bootstraps. Spatial maps use normalized-input coordinates; JVP uses eight frozen directions. Curves describe fixed checkpoints; depth contrasts use each model's family frozen before outcome access.

6.5 RQ3: Residual identity and branch

Figure 9 shows the two residual additions for one class target. Identity uses different input sites at the two additions and the cotangent at the corresponding addition output.

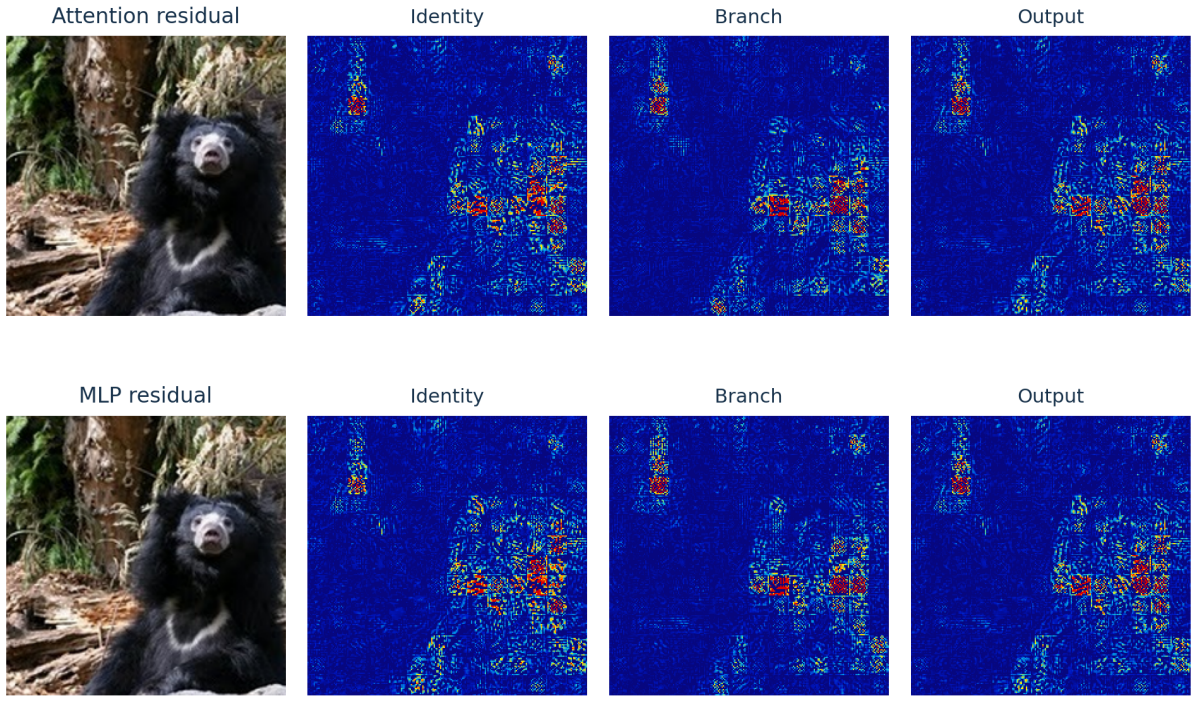


Figure 9. Rows show the attention and MLP residuals in DeiT-S/16 block 5 for one sloth-bear image and ground-truth logit. Columns show the input, identity/carry local map, branch contribution, and full-target Total Endpoint. Identity refers to different residual inputs in the two rows; complete signed maps satisfy identity plus branch equals Total Endpoint. Positive panels use independent P99 clipping and square-root scaling and are not additive. This single-image display has no statistical intervals.

Late-minus-early identity-relative-scale medians are attention -0.124 [-0.134, -0.116] / -0.045 [-0.049, -0.039]; MLP 0.180 [0.167, 0.189] / 0.178 [0.166, 0.188], with 400 images per model.

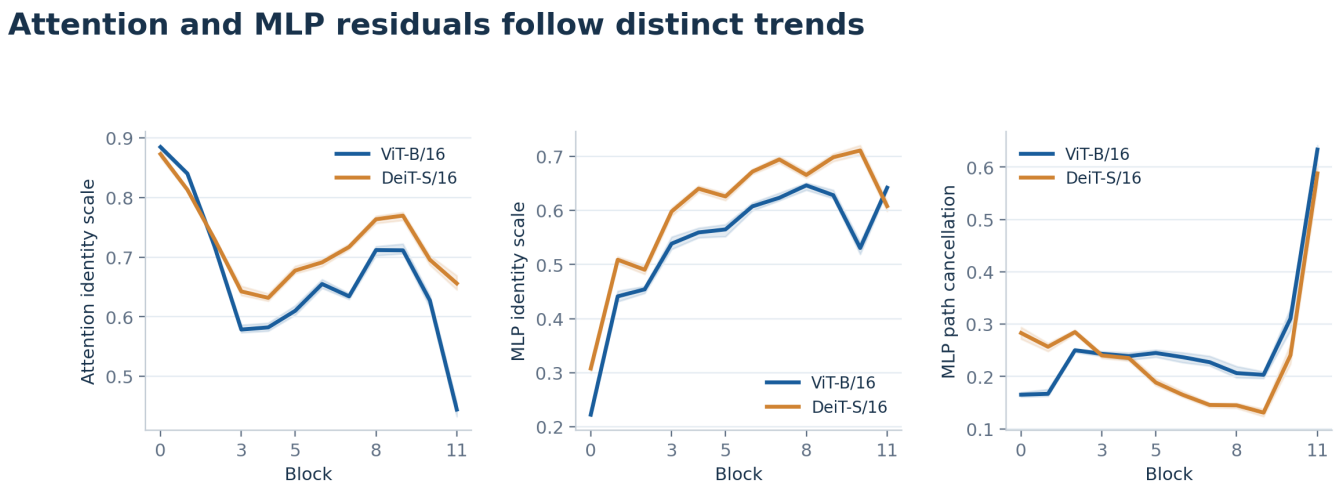


Figure 10. Left and middle panels show identity relative absolute mass for attention and MLP residuals. Right: path cancellation between signed MLP identity and branch maps. Each image contributes one value per residual and block; curves show medians across 400 images per checkpoint. Shading shows 95% intervals from 10,000 image bootstraps. All maps use normalized-input Endpoint. Curves describe fixed checkpoints; depth contrasts use model-specific families frozen before outcome access. The later output–identity spatial-TV analysis is separately post hoc.

The separately identified post-hoc spatial-TV analysis compares normalized output and identity patch mass. Late-minus-early medians and 95% intervals are vit_b16 attention 0.052 [0.049, 0.055]; vit_b16 mlp -0.019 [-0.022, -0.015]; deit_s16 attention 0.028 [0.026, 0.030]; deit_s16 mlp -0.046 [-0.052, -0.041]. Four tests pass their separate Holm family. A total map lacks identity assignment and an edge scalar lacks position, so this comparison needs both indices.

6.6 Independent-cohort and four-checkpoint robustness

The independent cohort tests four prospectively frozen hypotheses on 400 images with zero overlap with the original cohorts and the concurrent 500-image cohort. H1 is the block-11 minus block-0 entropy difference across 32 post-block neurons; H2 is the change in routing mass relative to Q/K-routing plus V mass; H3 subtracts the attention-residual identity trajectory from the MLP-residual identity trajectory; H4 is the path-map head-ranking correlation increment over activation × gradient, averaged across four representative blocks. Table 5 reports image-level medians and 95% bootstrap intervals.

Table 5. Independent-cohort four-checkpoint check; median [95% CI]

Checkpoint	H1	H2	H3	H4	Inference
vit_b16	0.413 [0.405, 0.420]	0.239 [0.221, 0.259]	0.851 [0.839, 0.865]	-0.114 [-0.128, -0.091]	400 images; fixed checkpoint; prospective robustness
deit_s16	0.378 [0.370, 0.383]	0.322 [0.310, 0.329]	0.522 [0.495, 0.543]	-0.186 [-0.229, -0.157]	400 images; fixed checkpoint; prospective robustness
vit_b16_in1k	0.336 [0.330, 0.340]	0.241 [0.218, 0.255]	0.632 [0.620, 0.648]	-0.186 [-0.205, -0.168]	400 images; fixed checkpoint; prospective robustness
vit_s16_augreg	0.390 [0.384, 0.399]	0.276 [0.256, 0.296]	0.867 [0.840, 0.891]	-0.136 [-0.171, -0.114]	400 images; fixed checkpoint; prospective robustness

H4 uses the ground-truth logit and normalized-zero reference. For image j, block b, and head h, let z_{jbh} denote the head-channel slice at the input to `attn.proj`, and define

$$M_{jbh} = \sum_i |\mathcal{L}_{jbh,i}|, \qquad G^{\mathrm{AG}}_{jbh} = |\langle z_{jbh}, \nabla_{z_{jbh}} s_j \rangle|, \tag{20}$$

$$Y_{jbh} = |s_j(x) - s_j(x; z_{jbh} \leftarrow z^{\text{normalized-zero}}_{jbh})|, \qquad H4_j = \frac{1}{4} \sum_{b \in \{0,3,7,11\}} [\rho_h(M_{jbh}, Y_{jbh}) - \rho_h(G^{\mathrm{AG}}_{jbh}, Y_{jbh})]. \tag{21}$$

G^{AG} takes the inner product over the complete head tensor before the absolute value; it is neither a sum of elementwise absolute values nor the attention probability A. Spearman uses average ranks. Fewer than four valid heads, nonfinite values, or a constant vector yield NA. The two correlations are differenced within heads at each block, followed by an arithmetic mean over blocks 0, 3, 7, and 11. The final median and bootstrap interval use 400 images; the two-sided exact sign test enters the four-test H1–H4 Holm family within each checkpoint.

H1–H3 are positive at all four checkpoints, with within-checkpoint Holm-adjusted $p < 10^{-6}$. H4 has the opposite direction at every checkpoint: within-image head-ranking correlation is 0.114–0.186 lower for path maps than for activation × gradient, with Holm-adjusted $p < 10^{-6}$. The spatial-broadening, routing, and residual-trajectory directions remain aligned on the independent image cohort and additional public checkpoints; stable path-map head-ranking gain is not supported. The added weights are not independent training seeds.

Dual indexing increases structural resolution and auditability. H4 does not outperform activation × gradient, so additional structure does not automatically improve finite-intervention prediction.

6.7 Coordinate parameterization and controls

This revision compares normalized-input Endpoint, raw-pixel gradient, and RGB-zero-origin weighted gradient at the same input point. Coverage is 100 fixed-unit images and 400 structural images per model, paired by image. All 48 phase-summary groups have complete image coverage and consistent main directions. For B11−B0, however, DeiT-S Q medians are −0.004, 0.003, and 0.002; all three intervals cross zero and point-estimate signs change. All 96 median differences, intervals, and directions appear in Supplement S17. Main conclusions remain restricted to normalized-input Endpoint, with object-specific direction retention reported separately as robustness evidence.

Table 5a. Eight core coordinate results; late-minus-early median difference (ViT-B / DeiT-S). The columns are normalized-input Endpoint, raw-pixel gradient, and zero-origin raw-pixel weighted gradient; they are not three forms of Endpoint.

Object / metric	Normalized-input Endpoint	Raw-pixel gradient	Zero-origin raw-pixel weighted gradient
Unit entropy	0.220 / 0.225	0.240 / 0.245	0.217 / 0.221
Unit own-token locality	-0.356 / -0.342	-0.367 / -0.359	-0.356 / -0.356
Unit normalized distance	0.131 / 0.122	0.138 / 0.129	0.134 / 0.127
Q entropy	0.037 / 0.018	0.039 / 0.018	0.034 / 0.015
K entropy	-0.021 / -0.028	-0.019 / -0.029	-0.019 / -0.027
V entropy	-0.010 / -0.013	-0.009 / -0.016	-0.010 / -0.015
Attention output–identity TV	0.052 / 0.028	0.054 / 0.028	0.053 / 0.027
MLP output–identity TV	-0.019 / -0.046	-0.017 / -0.044	-0.018 / -0.039

Table 5a reports the phase contrast. The coordinate-sensitive DeiT-S Q exception uses the distinct B11−B0 estimand, so the two contrasts are not interchangeable.

Trained versus random initialization, GT versus alternative targets, forward-feature matching, target-stratum/image resampling, and visual method comparisons are collected in the supplementary experiments. Random parameters also exhibit broadening; the parameter-state comparison does not identify a causal effect of the training process. Cross-layer and all-head regional replacements remain supplementary, separate from the original locally prespecified block-5 analysis.

7. Ranking in Prescribed Finite-Replacement Tasks

7.1 RQ4: Prescribed component replacement and region-induced designated-head activation replacement

The original 12 pooled component families remain available as frozen provenance, with continuous correlations, intervals, and Holm-adjusted p values in the supplement. Eleven of twelve met the original composite criterion, while the all-family claim remained unsupported. Table 6 instead prioritizes post-hoc A/V and attention/MLP stratified continuous estimates and type/depth mixing diagnostics, avoiding a single summary for distinct operations.

Table 6. Component association by type and depth diagnostic; image-level median [95% CI]

Model	Reference	A-only	V-only	Attention branch	MLP branch	Type+block-adjusted A/V	Type+block-adjusted residual
vit_b16	normalized_zero	0.049 [0.007, 0.091]	0.224 [0.192, 0.259]	-0.028 [-0.056, -0.003]	0.287 [0.266, 0.318]	0.030 [-0.003, 0.071]	-0.128 [-0.164, -0.084]
vit_b16	blur_rgb	0.073 [0.028, 0.112]	0.311 [0.266, 0.350]	0.000 [-0.021, 0.042]	0.378 [0.336, 0.402]	-0.018 [-0.063, 0.016]	0.042 [-0.007, 0.083]
deit_s16	normalized_zero	0.091 [0.049, 0.119]	0.035 [-0.021, 0.087]	-0.157 [-0.189, -0.119]	0.378 [0.346, 0.406]	0.089 [0.056, 0.120]	-0.091 [-0.120, -0.059]
deit_s16	blur_rgb	0.101 [0.056, 0.133]	0.147 [0.112, 0.182]	-0.024 [-0.070, 0.014]	0.371 [0.332, 0.395]	0.095 [0.055, 0.128]	0.142 [0.096, 0.176]

A-only, V-only, attention-branch, and MLP-branch correlations are computed across 12 blocks within each image. The last two columns separately rank the 24 predictor and effect observations and then adjust both for operation and

block; all results are post-hoc diagnostics. A/V directions and magnitudes depend on model and reference. MLP branches drive the pooled residual association, whereas attention-branch intervals include zero in three settings or are negative. After joint type and block adjustment, correlations are near zero or negative for both ViT-B references and change direction across references for DeiT-S. The historical pooled correlation therefore describes a mixed family and does not establish uniform within-type ranking for A, V, attention, or MLP operations. Supplement S18 reports every block, one-factor adjustment, Fisher-z head aggregation, A/V concordance, scale diagnostics, and pooled provenance.

In the supplementary within-block head comparison, Taylor achieves higher Spearman and Capture and lower Regret (Table 7). Across-layer residual identity has Endpoint-minus-Taylor Spearman differences of 0.086–0.421 but a zero median Capture@1 difference, indicating task- and metric-dependent incremental value.

Table 7. Within-block head ranking; median of image means [95% CI]

Model	Reference	Score	rho	Capture@1	Capture@3	Regret@1	Inference
vit_b16	normalized_zero	local_map	0.281 [0.272, 0.297]	0.517 [0.504, 0.535]	0.615 [0.609, 0.625]	0.170 [0.161, 0.182]	400 images; fixed heads; exploratory
vit_b16	normalized_zero	taylor	0.674 [0.666, 0.684]	0.804 [0.796, 0.812]	0.857 [0.851, 0.862]	0.057 [0.052, 0.062]	400 images; fixed heads; exploratory
vit_b16	blur_rgb	local_map	0.273 [0.259, 0.286]	0.507 [0.495, 0.520]	0.618 [0.610, 0.626]	0.116 [0.112, 0.123]	400 images; fixed heads; exploratory
vit_b16	blur_rgb	taylor	0.686 [0.676, 0.694]	0.820 [0.808, 0.835]	0.866 [0.860, 0.871]	0.035 [0.032, 0.037]	400 images; fixed heads; exploratory
deit_s16	normalized_zero	local_map	0.195 [0.176, 0.214]	0.561 [0.546, 0.574]	0.712 [0.705, 0.721]	0.238 [0.208, 0.266]	400 images; fixed heads; exploratory
deit_s16	normalized_zero	taylor	0.610 [0.600, 0.629]	0.803 [0.796, 0.812]	0.889 [0.882, 0.895]	0.086 [0.079, 0.092]	400 images; fixed heads; exploratory
deit_s16	blur_rgb	local_map	0.148 [0.133, 0.162]	0.551 [0.539, 0.566]	0.710 [0.702, 0.716]	0.133 [0.117, 0.146]	400 images; fixed heads; exploratory
deit_s16	blur_rgb	taylor	0.624 [0.610, 0.638]	0.843 [0.835, 0.852]	0.897 [0.893, 0.902]	0.044 [0.039, 0.048]	400 images; fixed heads; exploratory

The region-induced designated-head activation replacement experiment fixes three heads in block 5, a shared 4×4 region pool and a maximum 25% area. Figure 11 and Tables 8–10 report paired differences, absolute AUC and normalized regret. Paired median Endpoint-minus-regional-Taylor differences are −0.023, −0.024, −0.030 and −0.036. None of the four families meets the +0.05 criterion over every baseline. The candidate-set envelope maximizes effects only over evaluated sets and is not a global optimum over all regions. The clean-only local map shows no stable or practically useful ranking gain in this task. Taylor performs better but uses actual displacement, has a different information budget, and retains a gap to the candidate-set envelope.

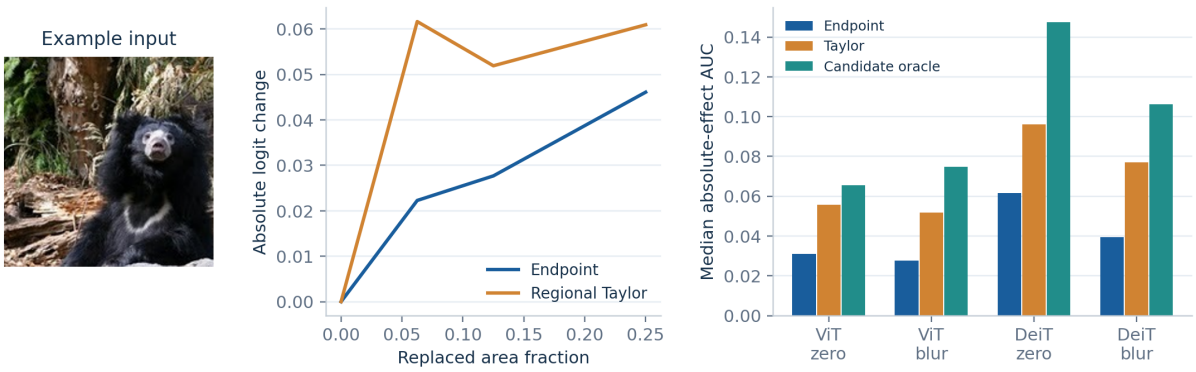


Figure 11. Left: the DeiT-S/16 sloth-bear input. Middle: mean absolute isolated effects across three fixed block-5 heads under normalized-zero reference, at replaced areas 0, 1/16, 2/16, and 4/16. Right: image-level median absolute-effect AUC for four model-by-reference families, each n=400, after averaging the three heads within images. Bars have no uncertainty bands; Tables 8–10 report 95% intervals. This is a descriptive display of fixed tasks. The candidate envelope takes the best evaluated effect separately at each area and is not a single globally optimal policy.

Table 8. Local-map regional AUC minus baseline; median difference [95% CI]

Model	Reference	vs Total Endpoint regional score	vs normalized-coordinate gradient	vs reference-aware Taylor	vs random	Original composite	Inference
vit_b16	normalized_zero	-0.001 [-0.002, -0.0002]	-3.1×10^{-5} [-0.001, 0.0003]	-0.023 [-0.026, -0.022]	-0.002 [-0.003, -0.001]	Unmet	400 paired images; fixed 3 heads; descriptive intervals
vit_b16	blur_rgb	-0.001 [-0.002, -0.0002]	-0.001 [-0.002, 3.1×10^{-5}]	-0.024 [-0.027, -0.022]	-0.003 [-0.005, -0.002]	Unmet	400 paired images; fixed 3 heads; descriptive intervals
deit_s16	normalized_zero	-0.002 [-0.003, 0.000]	-0.001 [-0.002, 0.001]	-0.030 [-0.034, -0.026]	0.001 [-0.001, 0.003]	Unmet	400 paired images; fixed 3 heads; descriptive intervals
deit_s16	blur_rgb	0.0002 [-0.0004, 0.001]	-0.0002 [-0.001, 0.0005]	-0.036 [-0.039, -0.031]	-0.001 [-0.002, 0.001]	Unmet	400 paired images; fixed 3 heads; descriptive intervals

Table 9. Absolute-effect AUC and logit; median [95% CI]

Model	Reference	Local-map regional AUC	Taylor AUC	Random AUC	Candidate-envelope AUC	Clean GT logit	Inference
vit_b16	normalized_zero	0.031 [0.029, 0.032]	0.056 [0.053, 0.058]	0.034 [0.031, 0.036]	0.065 [0.063, 0.070]	8.899 [8.842, 8.942]	400 images; fixed 3 heads; descriptive

Model	Reference	Local-map regional AUC	Taylor AUC	Random AUC	Candidate-envelope AUC	Clean GT logit	Inference
vit_b16	blur_rgb	0.028 [0.026, 0.029]	0.052 [0.049, 0.055]	0.031 [0.028, 0.033]	0.075 [0.068, 0.079]	8.899 [8.841, 8.943]	400 images; fixed 3 heads; descriptive
deit_s16	normalized_zero	0.062 [0.056, 0.068]	0.096 [0.087, 0.103]	0.061 [0.057, 0.065]	0.147 [0.137, 0.160]	8.797 [8.691, 8.856]	400 images; fixed 3 heads; descriptive
deit_s16	blur_rgb	0.039 [0.035, 0.043]	0.077 [0.068, 0.082]	0.041 [0.038, 0.044]	0.106 [0.098, 0.114]	8.797 [8.691, 8.856]	400 images; fixed 3 heads; descriptive

Table 10. Candidate-envelope regret and logit ratios; median [95% CI]

Model	Reference	Local-map regional regret	Taylor regret	Random regret	Local-map AUC/abs(logit)	Taylor AUC/abs(logit)	Inference
vit_b16	normalized_zero	0.530 [0.508, 0.555]	0.144 [0.132, 0.155]	0.506 [0.494, 0.524]	0.004 [0.003, 0.004]	0.006 [0.006, 0.007]	400 images; fixed 3 heads; descriptive
vit_b16	blur_rgb	0.637 [0.607, 0.655]	0.256 [0.244, 0.280]	0.593 [0.585, 0.600]	0.003 [0.003, 0.003]	0.006 [0.006, 0.006]	400 images; fixed 3 heads; descriptive
deit_s16	normalized_zero	0.565 [0.540, 0.586]	0.335 [0.317, 0.357]	0.595 [0.583, 0.601]	0.007 [0.006, 0.008]	0.011 [0.010, 0.012]	400 images; fixed 3 heads; descriptive
deit_s16	blur_rgb	0.629 [0.600, 0.640]	0.271 [0.249, 0.295]	0.615 [0.607, 0.624]	0.005 [0.004, 0.005]	0.009 [0.008, 0.010]	400 images; fixed 3 heads; descriptive

With the corrected random estimator, paired Endpoint-minus-random medians are −0.002, −0.003, 0.001 and −0.001; both DeiT-S intervals include zero. All four Taylor-minus-random intervals are positive. The block-5 three-head task follows the original locally prespecified protocol; the supplementary cross-layer all-head analysis covers blocks 2/5/10 and favors Taylor in 11/12 full-input and 12/12 isolated-head families (Supplement S14).

7.2 Behavioral calibration and estimand boundary

Cross-reference analysis clarifies this distinction. Endpoint uses the clean observed-input map, whereas Taylor explicitly uses replacement displacement. Across 163,200 component pairs, Taylor magnitude and absolute effect change under the original atol=1e-12, rtol=0 criterion. Their distributions and head-rank concordance are reported in supplementary tables. Reference choice changes intervention-related ordering, so local input maps and displacement estimators serve different questions.

8. Discussion and Conclusion

8.1 Discussion

Dual-index maps permit direct comparison of spatial distributions for specified uses or differential groups. Total maps describe overall broadening, component scalars describe scale, and Q/K/V entropy and residual output–identity TV require both path labels and input positions. Global cuts include bypass; branch partitions reconstruct only attention contribution. Distinct reconstruction targets keep accounting and interpretation consistent.

Empirical conclusions concern normalized-input Endpoint for fixed checkpoints and sampled targets. Main directions of unit broadening, K/V entropy, and residual TV persist across the three tested input representations, while DeiT-S Q B11—B0 shows coordinate and interval sensitivity. Equivalent hidden-coordinate transformations can also change components while preserving total Endpoint. These maps therefore describe the local differential structure of a specified implementation.

H4 is lower than activation × gradient at all four checkpoints. Dual-index maps increase structural resolution and auditability without automatically improving finite-intervention prediction. Taylor directly uses replacement displacement and better ranks the prescribed regional task. Historical pooled component correlations are positive, but the v7 stratification shows that A/V and attention/MLP are not homogeneous families; type-and-block-adjusted correlations depend on model and reference. The original all-family claim remains unsupported, and the behavioral evidence is broad but non-universal.

Image intervals condition on fixed targets and checkpoints rather than neuron or architecture populations. The independent image cohort uses the same team and code; public-checkpoint differences do not identify a causal effect of the training process. Random parameters also exhibit spatial broadening. Positive displays aid spatial reading; numerical inference uses complete signed/absolute arrays. The historical ViT-B Path-IG calibration failure is retained, and no cross-model Endpoint–Path inference is made.

8.2 Method limitations

A current-point first-order Jacobian describes local response and does not capture nonlinear interactions under large displacements. Components depend on the specified computational graph, internal coordinates, and cut and are not generally implementation invariant. Small gradients, saturation, and normalization can change spatial-shape metrics; normalized TV is not functional importance. The empirical scope is ImageNet, patch-16 ViTs, fixed scalar targets, and prescribed replacement contracts. Vector targets require a prespecified scalar functional or coordinate-wise analysis; sequence models and other architectures were not evaluated. Differences in training, scale, and data across checkpoints are not controlled, so the causes of model heterogeneity remain unidentified.

8.3 Conclusion

Under exact current-point derivative propagation, a scalar NNS learning image equals Gradient × Input. The technical contribution is a target-conditioned dual-index representation, distinct branch/global closure contracts, and analytic and mutation audits for concrete-use errors. Normalized-input Endpoint on fixed sampled targets exhibits block-index-associated spatial profiles. Coordinate checks identify object-specific direction retention and DeiT-S Q sensitivity. Independent-cohort H1–H3 directions persist, while H4 head ranking does not outperform activation × gradient. Prescribed replacement association is broad but non-universal and the all-family claim remains unsupported. Greater structural resolution and auditability do not automatically yield finite-intervention prediction gains.

References and Supplementary Material

References follow below, and the supplement accompanies the manuscript.

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